

AI-Assisted Study of a Mixed-Signal RTL-to-GDSII Flow Using OpenLane

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Abstract—This report presents a study of the *vsdmixedsignalflow* repository, an open-source project demonstrating mixed-signal physical design using the SKY130 PDK and OpenLane. The repository integrates an analog multiplexer macro with digital control logic and illustrates major RTL-to-GDSII implementation stages. AI-assisted workflows were used to understand repository organization, design inputs, LEF/LIB abstractions, OpenLane configuration, floorplanning, placement, routing, and verification concepts.

I. OBJECTIVE

The objective of this task was to understand the fundamentals of mixed-signal physical design and investigate how analog and digital blocks are integrated within an open-source RTL-to-GDSII implementation flow using OpenLane, SKY130, and analog macro abstractions such as LEF and LIB.

II. REPOSITORY SUMMARY

The *vsdmixedsignalflow* repository demonstrates a mixed-signal design consisting of an analog multiplexer macro integrated with supporting digital logic. The project includes Verilog source files, LEF physical abstractions, timing libraries (LIB), Magic layout files, and OpenLane configuration scripts.

- **Analog Macro Integration:** Demonstrates how an analog multiplexer macro is incorporated into a digital RTL-to-GDSII implementation flow using LEF and LIB abstractions.
- **Physical Design Flow:** Illustrates floorplanning, placement, PDN generation, routing, verification, and final layout generation using OpenLane and SKY130.

III. AI-ASSISTED WORKFLOW

AI tools were used to accelerate repository understanding and workflow planning. The following activities were performed using AI assistance:

- Repository structure analysis
- Study LEF and LIB files
- Analyze OpenLane flow stages
- Understand analog macro integration
- Learn routing and verification concepts
- Organize documentation and observations

IV. EXPERIMENTS ATTEMPTED

The following investigations were performed:

- 1) Exploration of repository structure
- 2) Analysis of top-level Verilog modules
- 3) Study of LEF and LIB files
- 4) OpenLane configuration review
- 5) Floorplanning analysis
- 6) Placement and routing study
- 7) Layout observation
- 8) Documentation of design stages

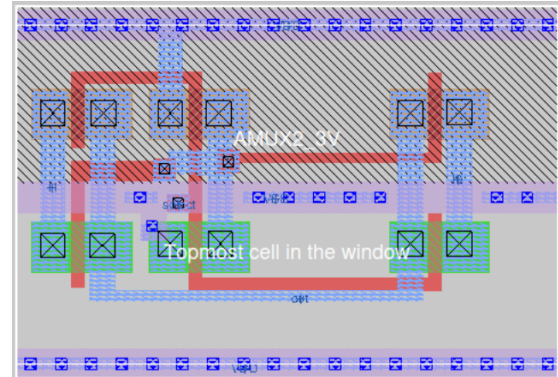


Fig. 1. Analog Multiplexer Layout

V. RESULTS AND OBSERVATIONS

The repository successfully demonstrates integration of an analog macro within a digital implementation environment. LEF and LIB representations allow the analog block to participate in physical design without exposing internal layout details.

- **Analog Macro Abstraction:** LEF and LIB views enable the analog multiplexer to be incorporated into the digital implementation flow while preserving internal layout confidentiality.
- **OpenLane-Based Implementation:** The OpenLane flow organizes implementation into synthesis, floorplanning, placement, power planning, routing, and verification stages, providing a practical example of mixed-signal design using open-source EDA tools.

VI. CHALLENGES AND NEXT STEPS

Challenges included understanding physical design terminology, LEF/LIB abstractions, OpenLane configuration parameters, and routing outputs. Future work includes executing OpenLane locally, reproducing flow stages, analyzing timing reports, performing DRC/LVS verification, and integrating additional analog macros.

VII. CONCLUSION

The study provided practical exposure to mixed-signal physical design concepts and open-source EDA tools. Analysis of the repository improved understanding of analog macro integration, OpenLane workflows, and physical design implementation stages, establishing a foundation for future mixed-signal design exploration.

REFERENCES

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- [3] SkyWater Technology and Google, “SKY130 open-source process design kit.”